MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY 12000252

TIME DELAY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the detail requirements for one type of circuit card assembly. The requirements of this specification together with the general requirements of specification MIL-C-50785 comprise the complete requirements for the Time Delay Circuit Card Assembly.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-50785	Circuit Card Assemblies, General Specification for

DRAWINGS

U.S. Army, Frankford Arsenal

12000239	Sight Current Generator
12000252	Circuit Card Assembly, Time Delay

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: (Commander, Frankford Arsenal, ATTN: SARFA-MDM, Phila., Pa. 19137) by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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(Copies of specifications, standards, drawings, and publications required by Suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Item definition.- This assembly is a single ended printed circuit board consisting of an operational amplifier and a ten second time delay circuit. The circuit card is used to amplify and invert inputs which make up portions of the magnet and torque current voltages, also control the ten second time delay relay (K23) when installed in the Sight Current Generator, Drawing Number 12000239.

3.1.1 Major component list.- The components of this assembly shall be as specified on the Engineering Parts List (EPL) 12000252.

3.1.2 Government furnished property list. Not applicable.

3.2 Characteristics.

3.2.1 Performance.- The assembly shall, when assembled in accordance with Drawing Number 12000252, be capable of performing the following functions.

a. Operational amplifier A1 shall amplify and invert one of the inputs required to develop the torque voltage.

b. Time delay circuit Q1, Q2 shall control relay K23 mounted in the sight current generator.

3.2.1.1 Input and output levels.

3.2.1.1.1 Input power.- The input power supplied shall be as specified in Table I under the conditions specified in Figure 1.

TABLE I - INPUT POWER

PIN	VOLTAGE
4 and 25	+15 ± 2.0 percent Vdc
14	-15 ± 2.0 percent Vdc
1 and 28	Ground

3.2.1.1.2 Output levels.

3.2.1.1.2.1 Pin 10 output.- The output shall be zero ± 50 millivolts between pin 10 and pin one with zero Vdc applied between pin 9 and pin one, and [Vin (-1.0)] ± 0.5 percent with a Vin of -13 Vdc to +13 Vdc applied between pin 9 and pin one.

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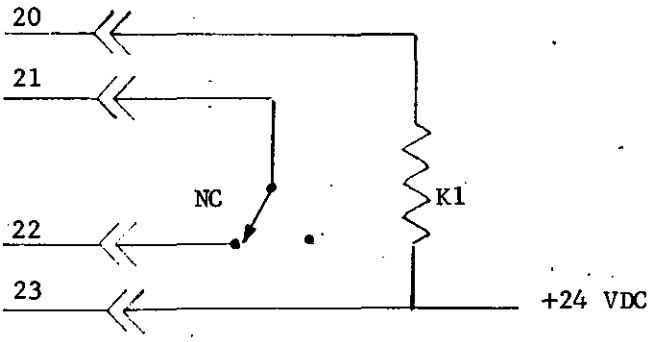


FIGURE 1. CIRCUIT DIAGRAM

3.2.1.1.2.2 Pin 20 output.- The voltage between pin 20 and pin 28 shall drop to zero Vdc ± 1.0 Vdc from 24 Vdc in 10 ± 4.0 seconds after 24 Vdc is applied between pin 23 and pin 28.

3.2.1.2 Environmental.- The assembly shall meet the environmental requirements of specification MIL-C-50785.

3.2.2 Physical characteristic.- The physical characteristics shall be as specified in drawing 12000252 and the engineering parts list.

3.3 Design and construction.

3.3.1 Production drawings.- This assembly shall be fabricated in accordance with the requirements of specification MIL-C-50785, drawing 12000252 and associated lists.

3.3.2 Standards of manufacture.- The standards of manufacture shall be those specified in specification MIL-C-50785.

3.3.3 Workmanship.- The workmanship shall be as specified in specification MIL-C-50785.

3.4 Preproduction sample.- Preproduction sampling shall be as specified in specification MIL-C-50785.

4. QUALITY ASSURANCE PROVISIONS

4.1 General.

4.1.1 Responsibility for inspection.- The responsibility for inspection shall be as specified in specification MIL-C-50785.

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4.1.2 Special tests and examination.- The assembly shall undergo first article and control testing in accordance with 4.1.2 of specification MIL-C-50785.

4.2 Quality conformance inspections.- The assembly shall be inspected and tested in accordance with 4.2 of specification MIL-C-50785.

4.2.1 Test equipment.

4.2.1.1 Power supply (PS_1).- The power supply shall be 15 ± 0.1 Vdc, 100 milliamperes or greater, ripple less than 10 millivolts (mv) rms or 100 mv peak-to-peak.

4.2.1.2 Power supply (PS_2).- The power supply shall be -15 ± 0.1 Vdc, 100 milliamperes or greater, ripple less than 10 mv rms or 100 mv peak-to-peak.

4.2.1.3 Power supply (PS_3).- The power supply shall be zero ± 13 Vdc, with accuracy of adjustment and regulation to ± .01 percent, 10 ma or greater.

4.2.1.4 Power supply (PS_4).- The power supply shall be 24 ± 0.5 Vdc, 100 ma or greater.

4.2.1.5 Relay (K1).- One relay, MS25329 - D1 or equivalent, is required.

4.2.1.6 Switch (S1).- One single pole, single throw switch is required.

4.2.1.7 Digital voltmeter.- The digital voltmeter (DVM) shall have a voltage accuracy and resolution of 0.01 percent or better from zero to 100 Vdc. Input impedance greater than 10 megohms on all voltage scales.

4.2.1.8 Oscilloscope.- The oscilloscope shall have voltage, time accuracy and resolution of 5.0 percent or better, with a rise time of 100 nanoseconds or better and with external triggering. Voltage sensitivity shall include the range from 0.05 volts/cm to 5.0 volts/cm.

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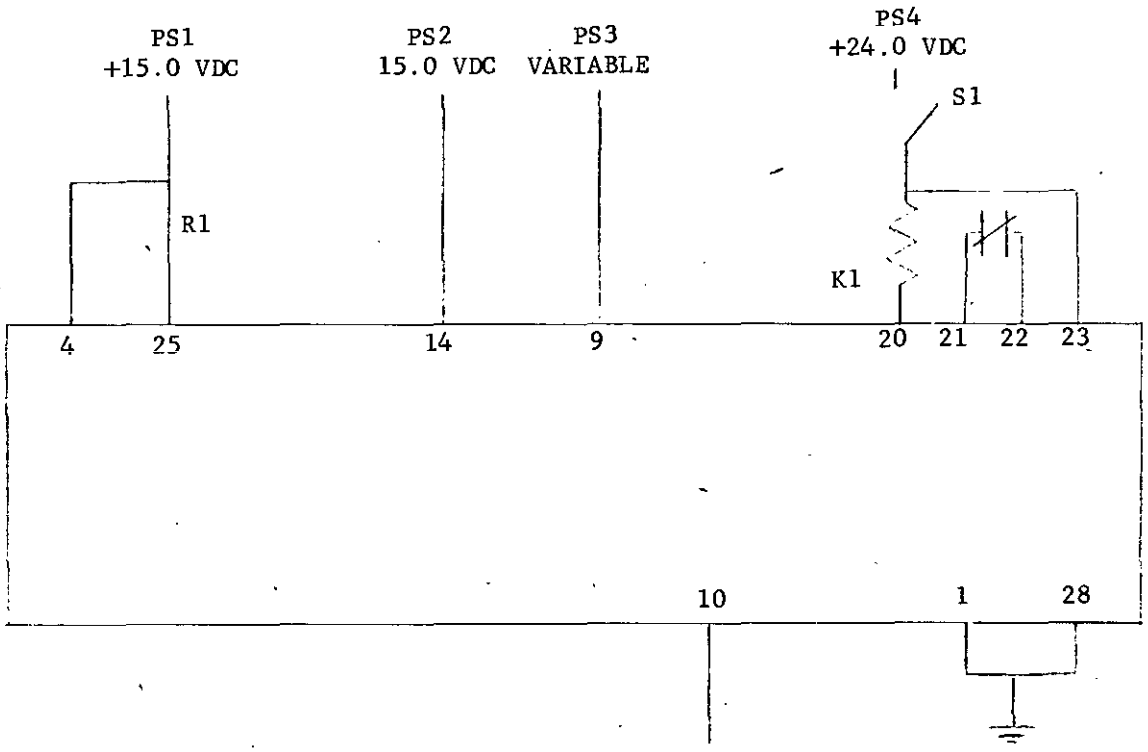


FIGURE 2. TEST CONNECTIONS

4.2.2 Functional tests.

4.2.2.1 Test connections.- The power supplies, resistors, and relay shall be connected to the assembly under test in accordance with Figure 2 using receptacle 10545455.

4.2.2.2 Test procedure.- Compliance with the requirements of 3.2.1.1.1 through 3.2.1.1.2.2 shall be verified as follows:

a. Power supply (PS_3) voltage shall be adjusted and connected to the specified input pin per Table II. A digital voltmeter shall be used to measure the voltage at the applicable output pin to the limits indicated in Table II.

b. The voltage at pin 20, as measured on the oscilloscope, shall be zero $\pm$ 1.0 Vdc, rising to +24 $\pm$ 2 Vdc when switch S1 is closed, then returning to zero $\pm$ 1.0 Vdc after a delay of 10 $\pm$ 4 seconds.

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TABLE II - TEST LIMITS

INPUT VOLTAGE (Vdc)	INPUT PIN	OUTPUT VOLTAGE (Vdc)	OUTPUT PIN
+13.0 ± 0.013	9	-13.0 ± 0.065	10
0 ± 0.010	9	0 ± 0.050	10
-13.0 ± 0.013	9	+13.0 ± 0.065	10

5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging and marking shall be in accordance with the level A requirements of Packaging Data Sheet (PDS), AMC Form 1029, 12000252.

6. NOTES

6.1 The notes specified in the general specification MIL-C-50785 are applicable to this specification.

Custodian:

Army - MU

Preparing Activity:

Army - MU

Project No. 1285-A456

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE
MIL-C-50785(MU) Circuit Card Assembly, Sight Current Generator Gen Spec For

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT/GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

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